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BATCH NORMALIZATION ORTHOGONALIZES REPRESENTATIONS IN DEEP RANDOM NETWORKS

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Abstract. This paper underlines a subtle property of batch-normalization (BN): Successive batch normalizations with random linear transformations make hidden representations increasingly orthogonal across layers of a deep neural network. We establish a non-asymptotic characterization of the interplay between depth, width, and the orthogonality of deep representations. More precisely, under a mild assumption, we prove that the deviation of the representations from orthogonality rapidly decays with depth up to a term inversely proportional to the network width. This result has two main implications: 1) Theoretically, as the depth grows, the distribution of the representation –after the linear layers– contracts to a Wasserstein-2 ball around an isotropic Gaussian distribution. Furthermore, the radius of this Wasserstein ball shrinks with the width of the network. 2) In practice, the orthogonality of the representations directly influences the performance of stochastic gradient descent (SGD). When representations are initially aligned, we observe SGD wastes many iterations to orthogonalize representations before the classification. Nevertheless, we experimentally show that starting optimization from orthogonal representations is sufficient to accelerate SGD, with no need for BN.

1. Introduction

Batch Normalization (BN) (Ioffe and Szegedy, 2015) enhances training across a wide range of deep network architectures and experimental setups (He et al., 2016; Huang et al., 2017; Silver et al., 2017). The practical success of BN has inspired research into the underlying benefits of BN (Santurkar et al., 2018; Karakida et al., 2019; Arora et al., 2019b; Bjorck et al., 2018). In particular, it shown that BN influences first-order optimization methods by avoiding the rank collapse in deep representation (Daneshmand et al., 2020), direction-length decoupling of optimization (Kohler et al., 2018), influencing the learning rate of the steepest descent (Arora et al., 2019b; Bjorck et al., 2018), and smoothing the optimization objective function (Santurkar et al., 2018; Karakida et al., 2019). However, the benefits of BN go beyond its critical role in optimization. For example, Frankle et al. (2020) shows that BN networks with random weights also achieve surprisingly high performance after only minor adjustments of their weights. This striking result motivates us to study the representational power of random networks with BN.

We study hidden representations across layers of a laboratory random BN with linear activations. Consider a batch of samples passing through consecutive BN and linear layers with Gaussian weights. The representations of these samples are perturbed by each random linear transformation, followed by a non-linear BN. At first glance, the deep representations appear unpredictable after many stochastic and non-linear transformations. Yet, we show that these transformations orthogonalize the representations. To prove this statement, we introduce the notion of "orthogonality gap", defined in Section 3, to quantify the deviation of representations from a perfectly orthogonal representation. Then, we prove that the orthogonality gap decays exponentially with the network

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depth and stabilizes around a term inversely related to the network width. More precisely, we prove

$$\mathbb{E}\left[\text{orthogonality gap}\right] = \mathcal{O}\left((1-\alpha)^{\text{depth}} + \frac{\text{batch size}}{\alpha\sqrt{\text{width}}}\right)$$

holds for $\alpha > 0$ that is an absolute constant under a mild assumption. In probability theoretic terms, we prove stochastic stability of the Markov chain of hidden representations (Kushner, 1967; Kushner and Yin, 2003; Khasminskii, 2011). The orthogonality of deep representations allows us to prove that the distribution of the representations after linear layers contracts to a Wasserstein-2 ball around isotropic Gaussian distribution as the network depth grows. Moreover, the radius of the ball is inversely proportional to the network width. Omitting details, we prove the following bound holds:

$$\text{Wasserstein}_2(\text{representations}, \text{Gaussian})^2 = \mathcal{O}\left((1-\alpha)^{\text{depth}}\,(\text{batch size}) + \frac{\text{batch size}}{\alpha\sqrt{\text{width}}}\right).$$

The above equation shows how depth, width, and batch size, interact with the law of the representations. Since the established rate is exponential with depth, the distribution of the representations stays in a Wasserstein ball around isotropic Gaussian distribution after a few layers. Thus, BN not only stabilizes the distribution of the representations, which is its main promise (Ioffe and Szegedy, 2015), but also enforces Gaussian isotropic distribution in deep layers.

There is growing interest in bridging the gap between neural networks, as the most successful parametric methods for learning, and Gaussian processes and kernel methods, as well-understood classical models for learning (Jacot et al., 2018; Matthews et al., 2018; Lee et al., 2019; Bietti and Mairal, 2019; Huang et al., 2014). This link is inspired by studying random neural networks in the asymptotic regime of infinite width. The seminal work by Neal (1996) sparks a single-layer network resembles a Gaussian process as its width goes to infinity. However, increasing the depth may significantly shift the distribution of the representations away from Gaussian (Ioffe and Szegedy, 2015). This distributional shift breaks the link between Gaussian processes and deep neural networks. To ensure Gaussian representations, Matthews et al. (2018) suggests increasing the width of the network proportional to the network depth. Here, we show that BN ensures Gaussian representations even for deep networks with *finite* width. This result bridges the link between deep neural networks and Gaussian process in the regime of finite width. Many studies rely on deep Gaussian representations in an infinite width setting (Yang et al., 2019; Schoenholz et al., 2017; Pennington et al., 2018; Klambauer et al., 2017; De Palma et al., 2019). Our non-asymptotic Gaussian approximation can be incorporated into their analysis to extend these results to the regime of finite width.

Since training starts from random networks, representations in these networks directly influence training. Hence, recent theoretical studies has investigated the interplay between initial hidden representations and training (Daneshmand et al., 2020; Bjorck et al., 2018; Frankle et al., 2020; Schoenholz et al., 2017; Saxe et al., 2014; Bahri et al., 2020). In particular, it is shown that hidden representations in random networks *without BN* become correlated as the network grows in depth, thereby drastically slows training (Daneshmand et al., 2020; He et al., 2016; Bjorck et al., 2018; Saxe et al., 2014). On the contrary, we prove that deep representations in networks *with BN* are almost orthogonal. We experimentally validate that initial orthogonal representations can save training time that would otherwise be needed to orthogonalize them. By proposing a novel initialization scheme, we ensures the orthogonality of hidden representations; then, we show that such an initialization effectively avoids the training slowdown with depth for vanilla networks, with no need for BN. This observation further motivates studying the inner workings of BN to replace or improve it in deep neural networks.

Theoretically, we made the following contributions:

(1) For MLPs with batch normalization, linear activation, and Gaussian i.i.d. weights, we prove that representations across layers become increasingly orthogonal up to a constant inversely proportional to the network width.
(2) Leveraging the orthogonality, we prove that the distribution of the representations contracts to a Wasserstein ball around a Gaussian distribution as the depth grows. Up to the best of our knowledge, this is the first *non-asymptotic* Gaussian approximation for deep neural networks with finite width.

Experimentally, we made the following contribution[1]:

(3) Inspired by our theoretical understanding, we propose a novel weight initialization for standard neural networks that ensure orthogonal representations without BN. Experimentally, we show that this initialization effectively avoids training slowdown with depth in the absence of BN.

2. Preliminaries

2.1. **Notations.** Akin to Daneshmand et al. (2020), we focus on a Multi-Layer Perceptron (MLP) with batch normalization and linear activation. Theoretical studies of linear networks is a growing research area (Saxe et al., 2014; Daneshmand et al., 2020; Bartlett et al., 2019; Arora et al., 2019a). When weights initialized randomly, linear and non-linear networks share similar properties (Daneshmand et al., 2020). For ease of analysis, we assume activations are linear. Yet, we will argue that our findings extend to non-linear in Appendix F.

We use n to denote batch size, and d to denote the width across all layers, which we further assume is larger than the batch size $d \geq n$. Let $H_\ell \in \mathbb{R}^{d\times n}$ denote n sample representations in layer ℓ, with $H_0 \in \mathbb{R}^{d\times n}$ corresponding to n input samples in the batch with d features. Successive representations are connected by Gaussian weight matrices $W_\ell \sim \mathcal{N}(0, I_d/d)$ as

$$H_{\ell+1} = \frac{1}{\sqrt{d}} BN(W_\ell H_\ell), \quad BN(M) = \text{diag}(MM^\top)^{-1/2} M, \tag{1}$$

where $\text{diag}(M)$ zeros out off-diagonal elements of its input matrix, and the scaling factor $1/\sqrt{d}$ ensures that all matrices $\{H_k\}_k$ have unit Frobenius norm (see Appendix 2). The BN function in Eq. (1) differs slightly from the commonly used definition for BN as the mean correction is omitted. However, Daneshmand et al. (2020) showes this difference does not change the network properties qualitatively. Readers can find all the notations in Appendix Table 2.

2.2. **The linear independence of hidden representations.** Daneshmand et al. (2020) observe that if inputs are linearly independent, then their hidden representations remain linearly independent in all layers as long as $d = \Omega(n^2)$. Under technical assumptions, Daneshmand et al. (2020) establishes a lower-bound on the average of the rank of hidden representations over infinite layers. Based on this study, we assume that the linear independence holds and build our analysis upon that. This avoids restating the technical assumptions of Daneshmand et al. (2020) and also further technical refinements of their theorems. The next assumption presents the formal statement of the linear independence property.

Assumption $\mathcal{A}_1(\alpha, \ell)$. *There exists an absolute positive constant α such that the minimum singular value of H_k is greater than (or equal to) α for all $k = 1, \ldots, \ell$.*

The linear independence of the representations is a shared property across all layers. However, the representations constantly change when passing through random layers. In this paper, we mathematically characterize the dynamics of the representations across layers.

[1]The codes for the experiments are available at https://github.com/hadidaneshmand/batchnorm21.git